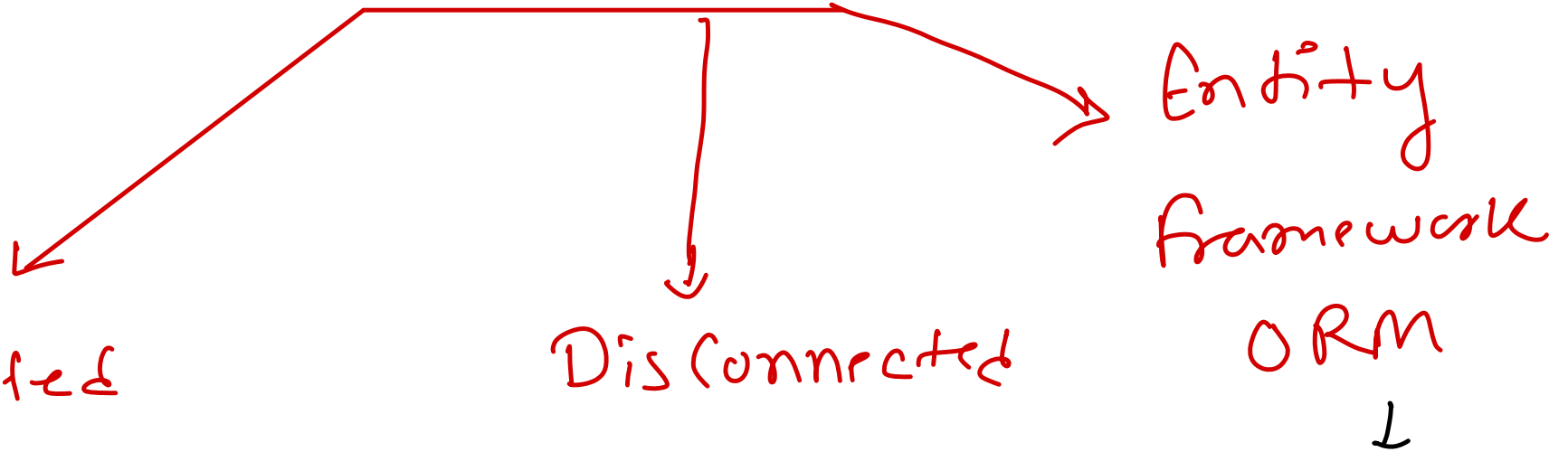


ADO.NET



Connected

Architecture

It is Developers responsibility to open and close the DbConnection manually

Disconnected

Architecture

Some other middleware layer / functionality
→ they connect & disconnect to Db on behalf of developer. So that this is no longer developer's responsibility

Microsoft SQL Server

* SQL Express edition :- free server for
learning & development
UN: sa, pwd: sa

* Standard SQL Server :- Paid $\begin{cases} \text{un} \\ \text{pwd} \end{cases}$

* Azure cloud :-

Visual Studio Editor

↳ .NET FW

↳ SQL Express } DB

↳ IIS Express

↳ Kestrel Server.

} web
server

OS → SQL Express → IIS Express →
.NET FW → Visual Studio

MySQL → SQL query

Oracle → PL-SQL query

MS SQL → T-SQL query

Connection string:-

"Server = (LocalDB)\MSSQLLocalDB;"

Initial Catalog = TESTDb;
Database = TESTDb

Integrated Security = true "

↓
Windows Authentication = true

Username = sa ; Password = sa "

System. Data. SqlClient.dll - Connected Arch.
Excel - OleDbClient.dll ODBCClient.dll
Old Database

MSSQLServer

SqlConnection con
class

class
SqlCommand cmd
cmd. ExecuteReader()

SqlDataReader reader
con. Open()
con. Close()

Object linking &
embedding [OLEDB]

OleDbConnection con

OleDbCommand cmd

OleDbDataReader reader

con. Open()

con. Close()

ODBC Connection

ODBC Command

ODBC DataReader

con. Open()

con. Close()

Old .NET →

System.Data.SqlClient.dll

.NET Core →

Microsoft.Data.SqlClient.dll